Israel Souza Ferreira

Machine Learning Researcher | ML Engineer

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PORTFOLIO

isrreal.github.io — projects, technical documentation, demos, and full resume.

PROFESSIONAL EXPERIENCE

Tieta Artificial Intelligence — **Remote, Brazil**
Machine Learning Researcher — CNPq RHAE Research Fellow — *Apr 2025 – Present*

- Developed **machine learning** models to investigate characteristics of educational questions and support assessment research.
- Contributed to experiments with textual and visual data, exploring multimodal and multitask approaches.
- Prototyped question-retrieval and ranking solutions to support analysis and research.
- Structured reproducible experiments, evaluations, and technical documentation to support the research cycle.

Contract Project — Educational Institution — **Remote, Brazil**
Machine Learning and Backend Developer — *2026 – Present*

- Independently delivered an end-to-end facial recognition attendance system currently running in production, covering machine learning, backend development, databases, security, testing, and deployment.
- Implemented 1:1 verification and 1:N facial identification using InsightFace, embeddings in $\mathbb{R}^{512}$, pgvector, *multi-frame* aggregation, *liveness* detection, and anti-replay protection.
- Trained an EfficientNet-B0 classifier for medical certificate validation, achieving approximately **85% accuracy** using Optuna, stratified validation, AMP, and a hybrid CNN + OCR decision pipeline.
- Built a FastAPI backend with JWT authentication, geofencing, PostgreSQL, SQLAlchemy, and Alembic; delivered Dockerized infrastructure with HTTPS, GitHub Actions, and **115 automated tests**.

Federal University of Ceará — Quixadá Campus — **On-site, Brazil**
Teaching Assistant for Algorithms, Calculus, and Pre-Calculus — *2022 – 2024*

- Provided academic support in data structures, algorithms, complexity analysis, and calculus applied to Computer Science.

SELECTED PROJECTS

- **Triple Roman Domination in Graphs:** first metaheuristic approaches reported in the literature for the problem, using a Genetic Algorithm, MAX-MIN Ant System, RVNS, and Integer Linear Programming for experimental validation.
- **Dengue Forecasting System:** MLOps pipeline under development for dengue case forecasting using Brazilian SINAN surveillance data, LSTM, PLE, PostgreSQL, MLflow, FastAPI, and Docker; reduced RMSE from approximately 98 to 83 for the Ceará time series.
- **Brazilian Emergency Aid — Query Benchmark:** system under redesign for benchmarking SQL indexing strategies over **257,170,290 records totaling 31.6 GB**, evaluating B-Tree and GIN/TRGM indexes across searches, aggregations, and joins using PostgreSQL, FastAPI, Streamlit, and Docker.

TECHNICAL SKILLS

Languages: Python, C++, SQL, and Bash.

Machine Learning: PyTorch, Scikit-learn, DSPy, FAISS, Optuna, InsightFace, OpenCV, Pandas, NumPy, and SciPy.

Backend, Data, and MLOps: FastAPI, PostgreSQL, pgvector, SQLAlchemy, Alembic, Streamlit, Docker, Git, GitHub Actions, GitLab CI/CD, MLflow, pytest, and mypy.

Methods: multimodal and multitask learning, MoE/MMoE, LSTM, attention, vector search, Item Response Theory, B-Tree and GIN/TRGM indexing, hypothesis testing, and ablation analysis.

EDUCATION

Federal University of Ceará — Quixadá Campus — **Brazil**
B.Sc. in Computer Science — *Expected Jul 2026*

LANGUAGES

Portuguese: Native | **English:** Fluent technical reading of research papers and documentation